

Benjamin Lewis

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Education

B.S. in Mechanical Engineering (GPA: 3.50)

Florida State University

Tallahassee, FL

Expected Graduation: May 2028

Minors: Physics, Mathematics, Innovation, General Business

Coursework: Mechanics of Materials, Mechanical Systems I, Engineering Design Methods, System Dynamics and Vibrations, Thermal Fluids I, Introduction to Electrical Engineering

Honors: Dean's List (2025)

Technical Skills

Applications: SolidWorks, AutoCAD, Onshape, VEXcode, Visual Studio Code, CLion

Programming: MATLAB, C++, Arduino, Python, R

Design and Analysis: Parametric and Feature-Based Modeling, Design for Additive Manufacturing, Kinematic Design, GD&T

Fabrication: CNC and manual machining (mills/lathes), drilling, tapping, reaming, sawing, grinding, sanding, soldering, welding, power and hand tools

Engineering Experience

Research Intern

High-Performance Materials Institute

Tallahassee, FL

Sep 2024 – Jan 2025

- Investigated the scalability of continuous carbon nanotube (CNT) yarn composites as high-performance alternatives to carbon fiber, contributing to a NASA-backed aerospace materials research team.
- Fabricated and cured three woven CNT panels using a manual loom; performed tensile testing to evaluate mechanical performance and fiber-matrix interaction.
- Quantified thermal effects on AroCy XU 371 cyanate ester resin density using a Mettler Toledo XS104 analytical balance; documented mass-volume measurements in Excel to inform curing process controls.
- Operated wet saws and sanding equipment to refine carbon fiber blocks to dimensional tolerances, supporting comparative analysis between CNT and conventional fiber architectures.

Work Experience

STEM Instructor

Young Engineers Tallahassee

Tallahassee, FL

May 2026 – Jul 2026, Sep 2026 – Present

- Co-instructed a 5-week STEM program with 4 other instructors, delivering hands-on curricula in maker tech, conservation engineering, biomedical technology, autonomous AI/robotics, and space systems.
- Mentored 15+ middle school students through a multi-week biomechatronics capstone project, directing the 3D printing of human brain models and the programming of Arduino sensor arrays to simulate neurological functions.
- Facilitated interactive embedded systems training using Arduino and Raspberry Pi, guiding 100% of students to successfully construct circuits and program custom logic loops for LEDs, motors, and sensors.
- Organized a final project showcase, enabling students to demonstrate their functional prototypes and present their design choices to peers, parents, and staff.

Engineering Projects

Elbow-Actuated Arm: Co-designed and delivered a custom elbow-assistive device through GRiP for a young patient with Type II Radial Aplasia using metal hardware, high-strength fishing line, and 3D-printed components.

Spring-Assisted Active Suspension: Modeled and animated a 2-DOF quarter-car suspension with angled spring-damper geometry in MATLAB using MotionGenesis equations of motion, while fabricating a K'NEX prototype for dynamic response testing.

3D Bioprinting Synthesis: Conducted a comprehensive literature synthesis of peer-reviewed sources, analyzing 3D bioprinting applications across organ fabrication, drug screening, and personalized healthcare.

Ergonomic Wheelchair Cane: Adapted a walking cane into a door-opening mechanism for wheelchair users, leveraging SolidWorks 3D modeling alongside decision matrices and Gantt charts for structured design planning.

Other

Certifications: Six Sigma Green Belt

Languages: Spanish – Professional Working Proficiency

Achievements: Merit badges in Programming, Robotics, and Welding

Institute of Industrial and Systems Engineers, Jan 2026

California State Seal of Biliteracy, May 2023

Boy Scouts of America, Aug 2020